

HALLUM, Sweden

WMO: 025420

Lat: 58.3218N

Lon: 13.0379E

Elev: 71

StdP: 100.47

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-12.9	-18.0	0.8	-16.1	-14.7	1.1	-12.5	13.4	6.3	12.2	5.3	0.6	190	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	26.9	18.6	25.0	17.8	23.1	16.9	19.7	24.6	18.8	23.4	17.8	21.9	2.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.1	21.5	17.1	12.3	20.6	16.1	11.5	19.5	56.6	24.5	53.6	23.6	50.4	21.9	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	7.8	DB	-19.9	29.4	5.1	1.6	-23.6	30.5	-26.6	31.5	-29.5	32.4	-33.2	33.5	(4)
(5)			WB	-19.8	21.2	4.9	0.9	-23.3	21.8	-26.2	22.3	-28.9	22.8	-32.4	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-1.9	-1.5	1.1	6.3	11.0	14.4	16.7	15.8	12.1	7.2	3.3	0.1		
	DBStd	7.58	5.09	4.73	4.52	3.31	3.46	2.77	2.67	2.45	2.80	3.60	3.74	5.07		
	HDD10.0	1736	368	322	276	118	28	1	0	1	12	100	202	307		
	HDD18.3	4128	626	556	534	361	228	121	68	84	188	346	452	566		
	CDD10.0	677	0	0	0	7	59	134	207	182	74	13	1	0		
	CDD18.3	28	0	0	0	0	1	4	16	7	1	0	0	0		
	CDH23.3	361	0	0	0	0	19	84	188	65	4	0	0	0		
	CDH26.7	53	0	0	0	0	1	11	34	6	0	0	0	0		
	WSAvg	3.2	3.7	3.3	3.3	3.2	3.1	2.9	2.7	3.0	3.2	3.2	3.4	3.6		
(15) Precipitation	PrecAvg	648	45	38	31	39	48	73	78	71	61	63	52	49		
	PrecMax	774	85	85	54	89	94	162	152	144	140	124	102	100		
	PrecMin	517	8	6	7	4	13	8	12	13	8	8	10	16		
	PrecStd	69	22	18	12	26	22	35	37	34	36	28	23	23		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	8.6	13.7	20.3	25.7	28.1	29.8	27.3	23.7	17.4	12.2	10.0		
		MCWB	6.9	7.0	8.4	12.5	16.1	18.6	19.4	18.8	17.0	14.2	11.2	9.2		
	2%	DB	6.6	6.8	10.8	17.8	23.0	25.9	27.4	25.1	20.6	14.7	10.8	8.6		
		MCWB	5.6	5.6	7.1	11.1	14.9	17.9	19.0	18.0	15.6	12.8	9.8	7.8		
	5%	DB	5.4	5.6	8.4	15.2	20.8	23.5	25.6	23.2	18.6	13.6	9.8	7.5		
		MCWB	4.5	4.8	5.7	9.6	14.1	16.9	18.6	17.3	14.8	12.2	9.0	6.6		
	10%	DB	4.3	4.5	6.6	12.7	17.9	21.2	23.3	21.2	17.1	12.4	8.6	6.2		
		MCWB	3.6	3.7	4.8	8.2	12.5	15.5	17.6	16.6	14.0	11.1	7.8	5.5		
	0.4%	WB	7.2	7.8	9.2	13.8	18.0	20.3	21.2	20.4	18.1	14.6	11.5	9.3		
		MCDB	7.9	8.6	12.7	19.5	23.4	25.5	25.9	24.3	21.6	16.3	12.0	9.7		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	5.8	5.9	7.5	11.6	16.2	18.6	20.0	19.2	16.2	13.4	10.0	7.9		
		MCDB	6.3	6.5	10.1	16.3	20.6	24.3	25.3	23.0	19.4	14.4	10.6	8.5		
	5%	WB	4.6	4.9	6.1	10.0	14.7	17.3	19.2	18.1	15.2	12.4	9.0	6.7		
		MCDB	5.2	5.5	7.8	14.4	19.4	22.6	24.3	21.7	18.0	13.4	9.7	7.4		
	10%	WB	3.6	3.7	5.0	8.7	13.4	16.1	18.3	17.1	14.3	11.3	7.9	5.5		
		MCDB	4.2	4.4	6.5	11.9	17.2	20.3	22.5	20.2	16.6	12.3	8.6	6.1		
	5% DB	MDBR	5.1	5.4	7.5	10.0	10.4	10.8	10.4	9.5	8.8	6.8	4.6	4.7		
		MCDBR	5.1	5.2	10.0	14.4	14.6	14.8	14.7	12.8	11.3	7.3	5.2	4.8		
		MCWBR	5.0	4.3	6.8	8.6	7.8	8.1	7.3	6.7	7.0	5.6	5.0	4.7		
		MCWBR	4.9	4.7	8.7	12.3	12.3	13.5	12.5	10.6	10.1	6.9	5.3	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.313	0.333	0.371	0.368	0.358	0.379	0.374	0.344	0.333	0.301	0.263		
		taud	2.240	2.315	2.375	2.329	2.397	2.447	2.407	2.420	2.492	2.443	2.315	2.176		
	Ebn at Noon		562	719	815	836	865	879	849	826	800	693	531	447		
		Edh at Noon	46	70	86	107	106	103	105	97	77	62	44	35		
(44) All-Sky Solar Radiation	RadAvg		0.35	0.92	2.43	3.91	5.15	5.76	5.16	4.07	2.78	1.26	0.45	0.24		
	RadStd		0.06	0.15	0.23	0.49	0.44	0.32	0.66	0.37	0.34	0.20	0.08	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page